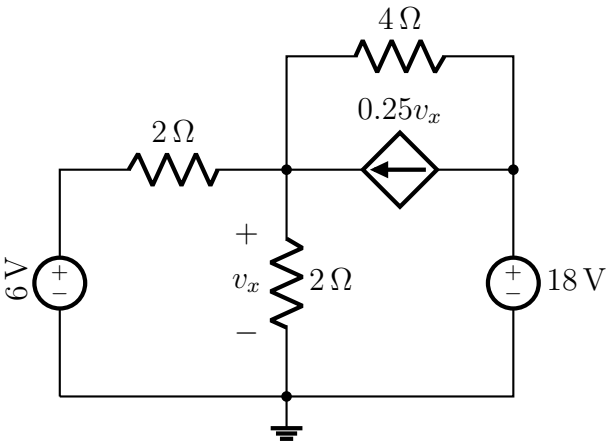
EE2015: Electric Circuits and Networks

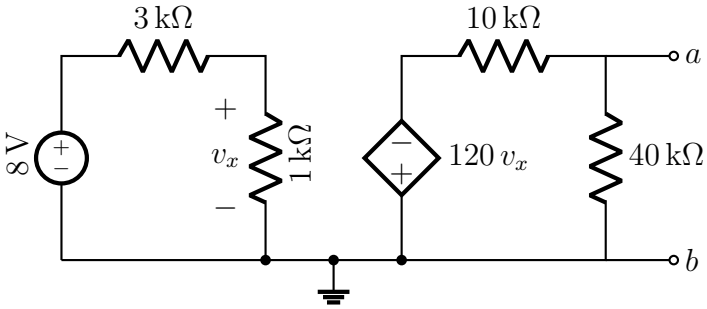
Tutorial 3

(August 22, 2025)

1. For the circuit shown on the right, find v_x by taking advantage of source transformation.



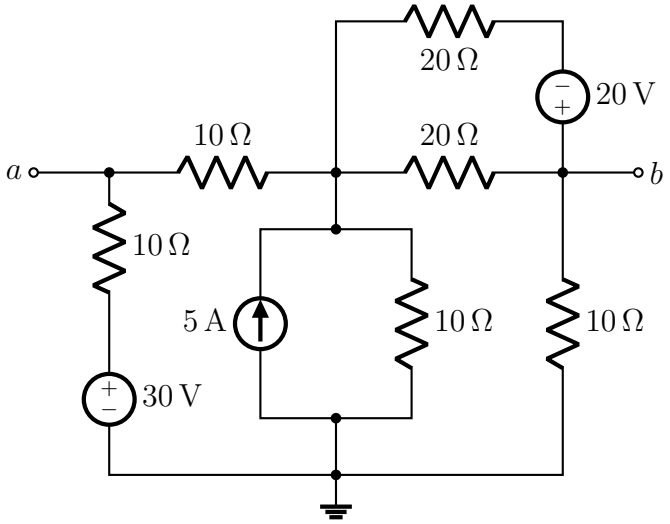
2.



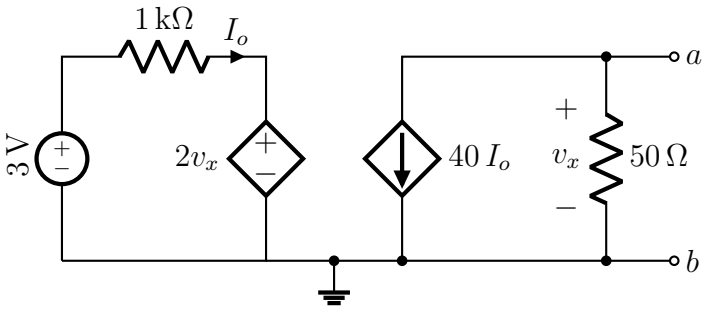
(a) For the circuit shown on the left, what resistor connected across the output terminals a-b will absorb maximum power from the circuit?

(b) What is the maximum value of power absorbed by the load?

3. For the circuit shown on the right, find the Norton equivalent as seen across terminals a-b.



4. For the circuit shown on the right, find the Thevenin equivalent as seen across terminals *a*-*b*.



5. An uknown ciruit has output terminals *a*-*b* across which a variable resistor R_L is connected. The load current I_L and load voltage V_L are tabulated below for three different values of R_L.

R_L (ohms)	V_L (volts)	I_L (amps)
2	6	3
8	16	2
14	21	1.5

   (a) Find I_L when $R_L = 4\,\Omega$.

   (b) What is the maximum power that this unknown circuit is capable of supplying to the load ?

6. In the circuit shown on the right, the 5 Ω load resistance is connected across *a*-*b*.

   (a) Find the Thevenin voltage $V_{\text{Th}}(s)$.

   (b) Find $Z_{\text{Th}}(s)$.

   (c) Find $v_L(0^+)$ and $v_L(\infty)$.

   (d) Find $v_L(t)$ for $t \geq 0$.

